

SI 313 WN26: Surveys Module Assignment

Part A: Survey Design

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Topic

AI in education - What are students' attitudes toward AI tools in their coursework? How do they use these tools, and what concerns do they have?

1. Research Objectives

With this survey, I'm trying to learn about student attitudes towards generative AI tools, focusing on the context of their coursework. Specifically, I'm focusing on University of Michigan students, since Michigan is uniquely positioned having its own genAI tools and partnerships with Google's genAI tools like Gemini and NotebookLM. GenAI is such a widely-discussed topic that limiting the scope to one university would make the results more focused, too. I'm most interested in figuring out how Michigan students currently use generative AI tools in their coursework, and what their perceptions are about the ethics of AI use in a university setting.

At this point in time, this question is obviously very relevant. As current University of Michigan students, we're at the inception of genAI tools in education. I personally have watched our use of these tools evolve so much in the past few years—I remember the first time my dad showed me ChatGPT as a senior in high school. Just under four years later, I think it's safe to say that every single one of my classmates is aware of these tools—if not extensively using them to assist their coursework. As students in 2026, how we learn with and use genAI tools will set a precedent for how they're used in universities down the line. Figuring out how students feel about these tools, how they use these tools, and what concerns they have about these tools is especially interesting and important. We can learn how genAI tools might need to adapt to the needs of students, and how other students might actually use these tools in the future.

2. Target Population and Sampling Strategy

My target population would be students at the University of Michigan Ann Arbor. The demographics of the students don't really matter for this study (besides age, only 18 and up since this research wouldn't need the thoughts of minors specifically), but I'd want students that have heard of popular genAI tools like ChatGPT. Specifically, I'd want students who have either used any genAI tools or seen them used by others twice within the past 14 days to get the most up-to-date sample possible. GenAI moves so fast, so having a current opinion is important here. Alternatively, I would include a question about whether students actively avoid genAI tools, so

even if they haven't interacted with the tools recently, their opinion can still be represented. I would focus on students across multiple colleges and academic disciplines to get the most representative sample possible as well.

I would recruit participants by sharing recruitment materials with friends and classmates who may be viable participants via text or email. Past that, I'd use the nomination/snowball recruitment technique and ask those friends and family to send information about my research to other U-M students in order to recruit the most possible participants. I would also try to get my survey sent out on large email lists like the Michigan Law school email list, any large student organizations like the Michigan Marching Band, or sororities/fraternities. Since I'm looking for opinions from all types of Michigan students, going to these larger areas is a great way to get a lot of participants quickly. I would also ask my professors if they'd be willing to send out information about these surveys to their class lists, especially in SI classes.

I might run into some issues with getting a representative sample. While convenience or snowball sampling is quick, it might get a lot of students in similar areas (majors, year in school, demographics, etc.) and skew the data around those groups. And while Greek life, the MMB, and MLaw are large, they aren't necessarily representative of all U-M students. The same goes for recruiting SI students through my classes. Also, the students that feel very strongly about genAI tools in either direction are probably more likely to respond than those in the middle, so that part of the population might be underrepresented. Finally, students might be hesitant to respond to questions about their personal genAI use.

3. Survey Questions

[Note: if publishing this survey, I would add a note at the very beginning stating that all responses are completely anonymous and won't be shared with U-M in order to ease social desirability bias that might come along with this topic, as well as any worries about things like being punished for AI use in class, etc...]

Q1 (Demographic): How old are you?

- Under 18 (TERMINATE)
- 18-24
- 25-34
- 35-44
- 45-54
- 55+

[Justification: Opinions of minors specifically aren't really relevant, so important to make this one of the first questions to get them out ASAP if they're under 18.]

Q2 (Demographic): Which U-M Ann Arbor school/college are you primarily enrolled in?

- Literature, Science, and the Arts (LSA)
- Ross School of Business
- College of Engineering
- School of Information (UMSI)
- School of Nursing
- School of Education
- Stamps School of Art & Design
- Taubman College of Architecture and Urban Planning
- School of Public Health
- Gerald R. Ford School of Public Policy
- School of Social Work
- Law School
- Medical School
- School of Dentistry
- College of Pharmacy
- School of Kinesiology
- School for Environment and Sustainability (SEAS)
- School of Music, Theatre & Dance (SMTD)
- Rackham Graduate School (Rackham)
- Other / Not listed
- Prefer not to say

[NOTE: This is the full list of schools/colleges, so there's a lot of options, which I recognize increases cognitive load. However, if I were actually running this survey, I would ideally want to use a text field with autocomplete and suggest school names based on the input, and not present all of these options to the user at once.]

[Justification: Different academic disciplines come with different coursework, instructor policies, and norms around the use of AI tools. This will be used to compare AI attitudes and use across students in different fields.]

Q3 (Demographic): What is your student level?

- Undergraduate
- Graduate/Master's
- Graduate/Professional (Law, Medicine, Dentistry, Pharmacy, etc.)
- Doctoral (PhD)
- Other
- Prefer not to say

[Justification: Like the question about what college the student is enrolled in, usage and norms might differ between different student levels. Undergraduates might not use or feel the same way about these tools that Doctoral students do.]

Q4 (Open-ended): In your own words, describe one way generative AI has helped you with your coursework. If you choose not to use generative AI for coursework, describe one reason why you haven't used it.

[Justification: Gets to more in-depth details and allows for any unexpected themes/ideas to pop up. Allows for both ends of the spectrum if they haven't used genAI for coursework.]

Q5 (Open-ended): Think about the most recent time you or someone you know considered using generative AI for coursework. What factors most influenced the decision to use or not use it?

[Justification: Gives an idea of the real reasons behind why students use generative AI or why they avoid it. It gets into their thought process. This might help explain any patterns seen in the closed ended questions.]

Q6 (Closed-ended): In the past 14 days, on how many days did you use a generative AI tool for coursework (e.g., ChatGPT, Gemini, Copilot, Claude, U-M tools)? Where "coursework" means any form of studying, assignments, labs, problem sets, projects, etc. done directly related to a U-M class.

- 0 days
- 1-2 days
- 3-4 days
- 5-6 days
- 7 days

[Justification: This establishes the baseline AI tool use and can segment non-users, light users, and heavy users.]

Q7 (Closed-ended): In the past 14 days, on how many days did you see other U-M students using generative AI tools for coursework (e.g., ChatGPT, Gemini, Copilot, Claude, U-M tools)?

- 0 days
- 1-2 days
- 3-4 days
- 5-6 days
- 7 days

[Justification: This captures the student's exposure to AI tools, which can affect the attitudes and even for those who don't use the tools themselves.]

Q8 (Closed-ended): Which statement about the use of generative AI tools for your coursework best describes you right now?

- I use generative AI for most of my coursework
- I often use it, but only for some classes or some of my coursework
- I rarely use it, but I've used it for my coursework before
- I don't use it right now, but I'm open to using it for my coursework
- I avoid using it for my coursework

[Justification: This identifies the stance towards AI tools and allows for that to be represented in analysis. Also includes the option about avoidance to cover for that case.]

Q9 (Closed-ended): For what coursework tasks have you used generative AI tools in the past 30 days? (Select all that apply)

- Brainstorming or ideation.
- Asking for explanations of concepts
- Summarizing readings or lecture notes
- Drafting or revising writing
- Coding help (writing/debugging code)
- Creating study materials (e.g. flashcards, practice questions, practice quizzes)
- Solving problem sets / showing solution steps
- Citation help (finding/formatting sources or making bibliographies)
- Group project support (planning, roles, slides)
- Other (please specify): _____
- I have not used generative AI for coursework in the past 30 days (please deselect all other options)

[Justification: This measures how students actually use AI tools, not just whether they use it. Enables us to find patterns of usage and potential links to how people perceive the benefits/risks of AI tools.]

Q10 (Closed-ended): When using generative AI for coursework, if at all, which tools do you primarily use? (Select all that apply)

- ChatGPT
- Google Gemini
- Google NotebookLM
- Microsoft Copilot
- Claude
- U-M AI tools (e.g., U-M GPT / Maizey)
- Other (please specify): _____
- I don't use generative AI for coursework

[Justification: This connects attitudes/concerns/etc. to specific tools. Can figure out if there are patterns between what tools people use and their thoughts about genAI]

Q11 (Closed-ended): How effective would you say using generative AI has been for completing your coursework, if at all?

- Extremely effective – I rely on generative AI; I would struggle to complete some coursework without it
- Very effective – It regularly improves my work and/or saves me time/effort
- Moderately effective – It helps frequently and improves my work and/or saves me some time
- Slightly effective – It helps occasionally, but it doesn't change my overall results much
- Not effective at all – It hasn't impacted my coursework
- I haven't used generative AI for coursework

[Justification: Captures the perceived value of genAI tools. This can be an outcome variable compared across different usage types, generative AI tools, and academic disciplines]

Q12 (Closed-ended): Please indicate how you feel about this statement: "Using generative AI in coursework is ethically acceptable as long as I follow course policies."

- Strongly agree – It's always ethically acceptable if I follow course policies
- Agree – It's usually ethically acceptable, with some exceptions
- Neither agree nor disagree – It's about equally acceptable / unacceptable
- Disagree – It usually isn't ethically acceptable, with some exceptions
- Strongly disagree – It's never ethically acceptable, even if I follow course policies

[Justification: Measures attitudes about the ethics of genAI use with coursework. Likert scale works the best here.]

Q13 (Closed-ended): What concerns, if any, do you have about students using generative AI in coursework? (Select all that apply)

- Cheating / academic integrity violations
- Over-reliance on generative AI tools that reduces learning
- Inaccurate or made-up information
- Bias or unfair outputs
- Privacy (what happens to my inputs)
- Equity (some students have better access or paid tools)
- Harder for instructors to assess learning
- Less original thinking / creativity
- None of the above

- Other (please specify): _____

[Justification: Specifies exactly what concerns students have after asking broadly about usage and attitudes earlier; supports more comparisons across different demographic groups.]

Q14 (Closed-ended) Overall, which statement best matches your general attitude toward the use of generative AI tools in coursework?

- Mostly positive — I'm excited/hopeful about its benefits
- Somewhat positive — I generally like it, with a few concerns
- Mixed/neutral — I see both pros and cons about equally
- Somewhat negative — I'm uneasy/worried about it overall
- Mostly negative — I'm fearful/pessimistic about its impacts

[Justification: captures a general sentiment about genAI tools in coursework, and allows segmenting of results by general attitude when analyzing the data]

4. Design Rationale

Question Wording

I avoided leading questions, double-barreled questions, and ambiguity when designing this survey. First of all, there are a lot of acronyms and colloquial terminologies used with these genAI tools (like the phrase “genAI” itself...) that I avoided using. I made sure to use the full names of these products and spell out “generative AI” to make the intention very clear throughout the survey design. I also defined what was meant by “coursework” to make that clear for participants so everyone was on the same page. I avoided leading questions by neutrally framing all of the questions, like asking about how often participants used genAI tools, what tools they used, and never implied whether AI was good, bad, ethical, unethical, etc. When asking about concerns or specific issues, I also spelled out that there might not be any in the question, like in Q11: “How helpful would you say generative AI has been for your coursework overall, if at all?” Lastly, I avoided double barreled questions by making sure even questions with multiple options were asking one key question; whether it be about what tools participants used or what concerns they might have. I also separated some questions about frequency of use like in Q6-Q7.

Response Options

For Q6-Q7, I chose these ordinal categories because it's easier for participants to recall a general range of times they used a genAI tool than the exact number of times. For Q9, Q10, and Q13, having the option to select multiple responses fits the wide variety of use of different AI tools, concerns about AI tools, and use cases for those tools that students across many

disciplines have. In Q11, Q12, and Q14, I added labels to each of the responses to remove the ambiguity of, for example, a simple “always, often, sometimes, rarely, never.” These scales would help responses be less random by giving users some context. I made these questions exhaustive and mutually exclusive too. I included “other (please specify)” and “prefer not to say” options where appropriate, and also made sure to check each choice to ensure that there wasn’t overlap between answers. I also made sure to include plenty of common scenarios surrounding genAI use. Most students would find a way that they use genAI in the answer choices.

Question Order

I started with eligibility and some demographic questions in Q1-Q3 to get a general idea about the participant. This also lets me have flexibility to terminate the survey quickly if they’re under 18. I started with open ended questions in Q4-Q5 capturing unprimed attitudes and stories from the participants. In Q6-Q10, I asked about behaviors, followed by attitudinal questions in Q11-14, ending with some open-ended reflection questions in Q4-5. This follows a general survey flow and funnel. I also considered some order effects. I moved around Q14 a bit, but ended up deciding that I wanted to ask about general attitudes around genAI use with coursework last. I wanted to ask participants all of the other questions first, so they could consider a fuller picture of the topic before answering. I originally had the open-ended questions at the end, but made sure to place the open-ended questions at the beginning, because those questions tire out users the quickest and because I want to capture the unprimed user view before priming them with other questions about topics like ethics and concerns around genAI tools.

Measurement Validity

In general I am confident that my questions measure what I intend them to measure. I want to look at U-M student use of genAI tools with their coursework, attitudes surrounding the use of those tools in coursework, and concerns surrounding the use of those tools in coursework, and I have questions directly measuring all of those things. Q6-Q10 all measure use, Q11-Q13 ask about attitudes, Q14 measures concerns, and Q4-Q5 both give open-ended data that helps interpret the patterns that might be seen in the rest of the data.

5. Potential Biases and Limitations

The largest response bias in this study would probably be social desirability bias. Especially if a professor were to send this out in an email, students might be afraid to answer questions about genAI usage because they might be afraid that professors would see the responses. While that recruiting technique is very solid for getting more participants, it might lead to that response bias. Also, the nature of genAI use is inherently sort of tricky, so there might be social desirability bias anyway. There might also be some moderacy bias here. AI and the use of genAI tools in schools is a touchy and complex subject, so some people might not have a fully developed opinion (or might not want to get into their full opinion) about that topic. Also, having multi-select questions with many options might cause users to satisfice and

potentially only pick the first option (primacy effect), even if it's important to the research and design to have those options

I anticipate there to be some coverage error here; the survey is going to be limited to students who come into contact with friends, student org lists, or certain classes. Students in smaller academic programs or those in other academic communities might not have a good chance of being sampled. I talked about potential challenges with getting a representative sample earlier, and that might lead to some sampling error if the sample isn't truly random or representative. Finally, there might be some nonresponse error based on when the survey is sent out in the school year. Students in different years of different programs might be busy at different times of year. Also, those worried about their responses about using AI tools being linked back to them might not want to respond to the survey.

My sampling strategy, while convenient, might limit generalizability because I'm essentially just connecting with my network, which might not be generalizable to the larger U-M student body. I tried to cover this for reaching out to a wide variety of students in different disciplines and organizations, but it wouldn't be perfect. This strategy might also recruit more people with similar interests to each other, and similar ideas about genAI usage and norms.

If I had more resources, I would work with the university directly to run the survey at a large scale, and actually take a random sample of U-M students to make it very generalizable. I might also sample smaller U-M social groups and apply weights to the survey results like the ANES does. I would offer incentives to respond (gift cards, money, etc., like the university or SI does for their surveys), send multiple reminders to potential participants, and have multiple modes of recruitment (email, in-class, people recruiting at the diag, etc.). In general, I would also make the survey a little larger and have some more open-ended questions if this were the case. If I had more resources, I would be able to gather more in-depth data, so I would want to make the most out of the opportunity.